

CG 1122

MATERIAL SAFETY DATA SHEET

The Valvoline Company

Page 001
Date Prepared: 06/19/06
Date Printed: 07/12/06
MSDS No: 525.0405333-002.001

LIQUID MACHINE COMP 4/1 GAL UNL

1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Material Identity

Product Name: LIQUID MACHINE COMP 4/1 GAL UNL
SAP Material No: CBOOB003-03-U

Wesmar Products Inc
1440 NW 45th St
Seattle, WA 98107
(206) 782-8186

Telephone Numbers
Emergency: 1-800-274-5263
Information: 1-859-357-7206

2. COMPOSITION/INFORMATION ON INGREDIENTS

Ingredient(s)	CAS Number	% (by weight)
KEROSENE, STRAIGHT RUN, KEROSENE	8008-20-6	15.0- 25.0
QUARTZ	14808-60-7	11.0- 21.0
MICROCRYSTALLINE SILICA	1317-95-9	11.0- 21.0
SILICA	61790-53-2	4.5- 15.0
PINE OIL	8002-09-3	1.0- 8.0
TALL OIL FATTY ACID	61790-12-3	1.0- 7.0
ALIPHATIC PETROLEUM DISTILLATES	8042-47-5	1.0- 6.0
MORPHOLINE	110-91-8	1.0- 6.0

3. HAZARDS IDENTIFICATION

Potential Health Effects

Eye

Material is corrosive to eyes. May cause burns. Additional symptoms of eye exposure may include: halo vision (blurred vision around bright objects).

Skin

Can cause skin burns and other permanent skin damage. Additional symptoms of skin contact may include: skin blistering, allergic skin reaction (delayed skin rash which may be followed by blistering, scaling and other skin effects), Passage of this material through the skin may be harmful.

Swallowing

Swallowing this material may be harmful or fatal. Symptoms may include severe stomach and intestinal irritation, burns and tissue damage. Shock may occur. This material can get into the lungs during swallowing or vomiting. This results in lung inflammation and other lung injury.

Inhalation

Breathing of vapor or mist is possible. Breathing this material may be harmful or fatal. May cause severe irritation and burns to the nose, throat, and respiratory tract. Prolonged or repeated breathing of this material may result in chronic bronchitis (inflammation of the airways of the lungs). Symptoms include coughing and shortness of breath. Symptoms usually occur at air concentrations higher than the recommended exposure limits (See Section 8). Another form of fibrosis, acute silicosis, can occur with exposures to very high concentrations of respirable silica over shorter periods of time, sometimes as short as a few months. Symptoms of acute silicosis include progressive shortness of breath, fever, cough and weight loss. Acute silicosis is fatal.

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Symptoms of Exposure

Signs and symptoms of exposure to this material through breathing, swallowing, and/or passage of the material through the skin may include: stomach or intestinal upset (nausea, vomiting, diarrhea), thirst, irritation (nose, throat, airways), runny nose, lung irritation, cough, central nervous system depression (dizziness, drowsiness, weakness, fatigue, nausea, headache, unconsciousness), muscle weakness, pain in the abdomen, loss of coordination, confusion, difficult breathing, irregular heartbeat, narcosis (dazed or sluggish feeling), lung edema (fluid buildup in the lung tissue), kidney damage, liver damage, lung damage, convulsions, coma, and death. Exposure to this product (or a component) may cause an allergic reaction (narrowing of the air passages of the lungs resulting in difficult breathing, tightness in the chest, coughing and wheezing) in some sensitive individuals. Other symptoms of an allergic reaction may include itchy and watery eyes, runny and stuffy nose, sweating, flushing, hives, rapid heart rate, and lowered blood pressure.

Target Organ Effects

Exposure to this material (or a component) has been found to cause kidney damage in male rats. The mechanism by which this toxicity occurs is specific to the male rat and the kidney effects are not expected to occur in humans. Chronic feeding studies with white oils (greater than 0.5% in the diet) have resulted in increased organ weights (liver, kidney, spleen), and in microscopic areas of oil accumulation in the liver and the lymph nodes of the gut in experimental animals. These changes have occasionally been associated with cell damage. In humans, cell damage resulting from accumulation of these oil droplets has only occurred following prolonged ingestion of mineral oil. These effects are not expected to occur as a result of occupational exposure to white oils. Overexposure to this material (or its components) has been suggested as a cause of the following effects in laboratory animals, and may aggravate preexisting disorders of these organs in humans: nasal damage, eye damage, kidney damage, liver damage, lung damage. Overexposure to this material (or its components) has been suggested as a cause of the following effects in humans, and may aggravate preexisting disorders of these organs: chronic bronchitis, emphysema, kidney damage.

Developmental Information

Tall oil fatty acids did not cause harm to the fetus when given in the diet of laboratory animals.

Cancer Information

This product (or a component) is a petroleum-derived material. Similar materials and certain compounds occurring naturally in petroleum oils have been shown to cause skin cancer in laboratory animals following repeated exposure without washing or removal. The International Agency for Research on Cancer (IARC) and the National Toxicology Program have determined that there is sufficient evidence in humans for the carcinogenicity of inhaled crystalline silica in the form of quartz or cristobalite. In addition, IARC has determined that there is sufficient evidence for the carcinogenicity of quartz and cristobalite in experimental animals. Among individuals with silicosis, lung cancer occurs more frequently in those who smoke. White mineral oil was not carcinogenic in laboratory animals by any route of exposure other than by injection into the abdomen. Results of abdominal injection are not relevant to possible human exposure to this material.

Other Health Effects

Nitrites should not be added to this material because this can result in formation of nitrosamines. Many nitrosamines cause cancer in laboratory animals. There is some evidence that breathing respirable crystalline silica or the disease silicosis is associated with an increased incidence of significant disease endpoints such as scleroderma (an immune system disorder manifested by fibrosis of the lungs, skin and other internal organs) and kidney disease.

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Primary Route(s) of Entry

Inhalation, Skin absorption, Skin contact, Eye contact, Ingestion.

4. FIRST AID MEASURES

Eyes

If material gets into the eyes, immediately flush eyes gently with water for at least 15 minutes while holding eyelids apart. If symptoms develop as a result of vapor exposure, immediately move individual away from exposure and into fresh air before flushing as recommended above. Seek immediate medical attention.

Skin

Immediately flush skin with water for at least 15 minutes while removing contaminated clothing and shoes. Seek immediate medical attention. Wash clothing before reuse and decontaminate or discard contaminated shoes.

Swallowing

Seek immediate medical attention. Do not induce vomiting. Vomiting will cause further damage to the mouth and throat. If individual is conscious and alert, immediately rinse mouth with water and give milk or water to drink. If possible, do not leave individual unattended.

Inhalation

If symptoms develop, immediately move individual away from exposure and into fresh air. Seek immediate medical attention; keep person warm and quiet. If person is not breathing, begin artificial respiration. If breathing is difficult, administer oxygen.

Note to Physicians

Inhalation of high concentrations of this material, as could occur in enclosed spaces or during deliberate abuse, may be associated with cardiac arrhythmias. Sympathomimetic drugs may initiate cardiac arrhythmias in persons exposed to this material. This material is an aspiration hazard. Potential danger from aspiration must be weighed against possible oral toxicity (See Section 3 - Swallowing) when deciding whether to induce vomiting. Acute aspiration of large amounts of oil-laden material may produce a serious aspiration pneumonia. Patients who aspirate these oils should be followed for the development of long-term sequelae. Repeated aspiration of small quantities of mineral oil can produce chronic inflammation of the lungs (i.e. lipoid pneumonia) that may progress to pulmonary fibrosis. Symptoms are often subtle and radiological changes appear worse than clinical abnormalities. Occasionally, persistent cough, irritation of the upper respiratory tract, shortness of breath with exertion, fever, and bloody sputum occur. Inhalation exposure to oil mists below current workplace exposure limits is unlikely to cause pulmonary abnormalities. Preexisting disorders of the following organs (or organ systems) may be aggravated by exposure to this material: respiratory tract, skin, lung (for example, asthma-like conditions), liver, kidneys, eye. Individuals with pre-existing heart disorders may be more susceptible to arrhythmias (irregular heartbeats) if exposed to high concentrations of this material. Silicosis predisposes the individual to the development of tuberculosis. This is most likely to occur after the age of 50 and in association with moderate to severe silicosis.

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5. FIRE FIGHTING MEASURES

Flash Point

> 200.0 F (93.3 C)

Explosive Limit

No data

Autoignition Temperature

No data

Hazardous Products of Combustion

May form: carbon dioxide and carbon monoxide, various hydrocarbons.

Fire and Explosion Hazards

Vapors are heavier than air and may travel along the ground or may be moved by ventilation and ignited by pilot lights, other flames, sparks, heaters, smoking, electric motors, static discharge, or other ignition sources at locations distant from material handling point. Never use welding or cutting torch on or near drum (even empty) because product (even just residue) can ignite explosively.

Extinguishing Media

regular foam, water fog, carbon dioxide, dry chemical.

Fire Fighting Instructions

Wear a self-contained breathing apparatus with a full facepiece operated in the positive pressure demand mode with appropriate turn-out gear and chemical resistant personal protective equipment. Refer to the personal protective equipment section of this MSDS.

NFPA Rating

Health - 3, Flammability - 2, Reactivity - 0

6. ACCIDENTAL RELEASE MEASURES

Small Spill

Eliminate all sources of ignition such as flares, flames (including pilot lights), and electrical sparks. Absorb liquid on vermiculite, floor absorbent or other absorbent material. Persons not wearing proper personal protective equipment should be excluded from area of spill.

Large Spill

Prevent run-off to sewers, streams or other bodies of water. If run-off occurs, notify proper authorities as required, that a spill has occurred. Persons not wearing protective equipment should be excluded from area of spill until clean-up has been completed. Eliminate all ignition sources (flares, flames, including pilot lights, electrical sparks).

7. HANDLING AND STORAGE

Handling

Containers of this material may be hazardous when emptied. Since emptied containers retain product residues (vapor, liquid, and/or solid), all hazard precautions given in the data sheet must be observed. All five gallon pails and larger metal containers including tank cars and tank trucks should be grounded and/or bonded when material is transferred. Warning. Sudden release of hot organic chemical vapors or mists from process equipment operating at

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elevated temperature and pressure, or sudden ingress of air into vacuum equipment, may result in ignitions without the presence of obvious ignition sources. Published "autoignition" or "ignition" temperature values cannot be treated as safe operating temperatures in chemical processes without analysis of the actual process conditions. Any use of this product in elevated temperature processes should be thoroughly evaluated to establish and maintain safe operating conditions.

Storage

Store in closed containers in a dry, well-ventilated area. Do not store near extreme heat, open flame, or sources of ignition.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Eye Protection

Chemical splash goggles and face shield (8" min.) in compliance with OSHA regulations are advised; however, OSHA regulations also permit other type safety glasses. (Consult your industrial hygienist.)

Skin Protection

Wear impervious gloves (consult your safety equipment supplier). To prevent skin contact, wear impervious full-body protective clothing. Other protective equipment: eyewash station, emergency shower.

Respiratory Protections

If workplace exposure limit(s) of product or any component is exceeded (See Exposure Guidelines), a NIOSH/MSHA approved air supplied respirator is advised in absence of proper environmental control. OSHA regulations also permit other NIOSH/MSHA respirators (negative pressure type) under specified conditions (consult your industrial hygienist). Engineering or administrative controls should be implemented to reduce exposure.

Engineering Controls

Provide sufficient mechanical (general and/or local exhaust) ventilation to maintain exposure below TLV(s).

Exposure Guidelines

Component

KEROSENE, STRAIGHT RUN, KEROSENE (8008-20-6)
ACGIH TLV 200.000 mg/m³ - TWA ((Skin))

QUARTZ (14808-60-7)
OSHA VPEL 0.100 mg/m³ - TWA respirable dust
ACGIH TLV 0.025 mg/m³ - TWA

MICROCRYSTALLINE SILICA (1317-95-9)
OSHA VPEL 0.100 mg/m³ - TWA respirable dust (as quartz)

SILICA (61790-53-2)
OSHA VPEL 6.000 mg/m³ - TWA
ACGIH TLV 3.000 mg/m³ - TWA respirable particulate
ACGIH TLV 10.000 mg/m³ - TWA inhalable particulate

PINE OIL (8002-09-3)
OTHER LIMIT 100.000 ppm - TWA

TALL OIL FATTY ACID (61790-12-3)
No exposure limits established

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ALIPHATIC PETROLEUM DISTILLATES (8042-47-5)
OSHA VPEL 5.000 mg/m³ - TWA as Oil Mist - CASRN 8012-95-1
ACGIH TLV 5.000 mg/m³ - TWA as Oil Mist - CASRN 8012-95-1
ACGIH TLV 5.000 mg/m³ - TWA for mineral oil mist
ACGIH TLV 10.000 mg/m³ - STEL as Oil Mist - CASRN 8012-95-1
ACGIH TLV 10.000 mg/m³ - STEL for mineral oil mist

MORPHOLINE (110-91-8)
OSHA VPEL 20.000 ppm - TWA ((Skin))
OSHA VPEL 30.000 ppm - STEL ((Skin))
ACGIH TLV 20.000 ppm - TWA ((Skin))
ACGIH TLV 30.000 ppm - STEL ((Skin))

9. PHYSICAL AND CHEMICAL PROPERTIES

Boiling Point
(for product) > 150.0 F (65.5 C) @ 760.00 mmHg

Vapor Pressure
(for component) 7.000 mmHg

Specific Vapor Density
No data

Specific Gravity
1.289 @ 60.00 F

Liquid Density
10.760 lbs/gal @ 60.00 F
10.760 lbs/gal @ 60.00 F

Percent Volatiles (Including Water)
No data

Evaporation Rate
No data

Appearance
No data

State
LIQUID

Physical Form
No data

Color
BROWN

Odor
PINE

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pH
No data

10. STABILITY AND REACTIVITY

Hazardous Polymerization

Product will not undergo hazardous polymerization.

Hazardous Decomposition

May form: carbon dioxide and carbon monoxide, various hydrocarbons.

Chemical Stability

Stable.

Incompatibility

Avoid contact with: acids, halogenated hydrocarbons, strong oxidizing agents.

11. TOXICOLOGICAL INFORMATION

Chronic/Carcinogenicity

One study with morpholine in laboratory animals produced cancer, while others have not. The tumors in the one study may have resulted from exposure to N-nitrosomorpholine, an animal carcinogen. N-nitrosomorpholine can occur as a contaminant in morpholine or as a result of the interaction of morpholine with nitrite of unknown origin. There is no evidence that morpholine causes cancer in humans.

12. ECOLOGICAL INFORMATION

No data

13. DISPOSAL CONSIDERATION

Waste Management Information

Dispose of in accordance with all applicable local, state and federal regulations. Do not discharge effluent containing this product into lakes, streams, ponds or estuaries, oceans, or other waters unless in accordance with the requirements of a National Pollutant Discharge Elimination System (NPDES) permit, and the permitting authority has been notified in writing prior to discharge. Do not discharge effluent containing this product to sewer systems without previously notifying the local sewage treatment plant authority. For guidance, contact your State Water Board or Regional Office of the EPA.

14. TRANSPORT INFORMATION

DOT Information - 49 CFR 172.101

DOT Description:

Not Regulated

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Container/Mode:

CASES/SURFACE - NO EXCEPTIONS

NOS Component:

None

RQ (Reportable Quantity) - 49 CFR 172.101

Not applicable

15. REGULATORY INFORMATION

US Federal Regulations

CERCLA RQ - 40 CFR 302.4

None

SARA 302 Components - 40 CFR 355 Appendix A

None

Section 311/312 Hazard Class - 40 CFR 370.2

Immediate(X) Delayed(X) Fire(X) Reactive() Sudden Release of
Pressure()

SARA 313 Components - 40 CFR 372.65

None

International Regulations

Inventory Status

Not determined

State and Local Regulations

California Proposition 65

The following statement is made in order to comply with the California Safe Drinking Water and Toxic Enforcement Act of 1986: This product contains the following substance(s) known to the state of California to cause cancer.

QUARTZ

MICROCRYSTALLINE SILICA

New Jersey RTK Label Information

KEROSENE	8008-20-6
SILICA, QUARTZ	14808-60-7
SILICA, TRIPOLI	1317-95-9
PINE OIL	8002-09-3
MORPHOLINE	110-91-8

Pennsylvania RTK Label Information

KEROSINE (PETROLEUM)	8008-20-6
QUARTZ (SiO2)	14808-60-7
TRIPOLI	1317-95-9
MORPHOLINE	110-91-8

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16. OTHER INFORMATION

The information accumulated herein is believed to be accurate but is not warranted to be whether originating with the company or not. Recipients are advised to confirm in advance of need that the information is current, applicable, and suitable to their circumstances.